Information Retrieval Scientific Article Challenge

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Idea: locate the reference sentence as query

- In this challenge, relevance is defined by **citation relationships**
- the relevant documents are the papers that the query paper **cites**

- TF - IDF input: **Full text** in parquet.
- Dense Retrieval input: **Full text** in npy.

- Citation part: abstract | introduction | methodology | dataset
- Will it be better to use **the exact reference sentence** as query?

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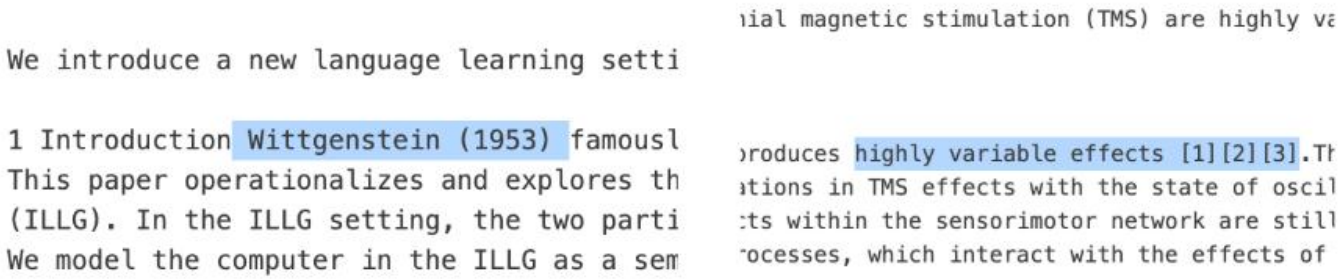
Pipeline:

1. build_corpus_index: Year - Author - Title - Abstract
2. temporal_filter: reference <= query
3. extract_author_refs: based on original text, mainly 2 kinds of citation



1.jpeg

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Pipeline:

1. build_corpus_index
2. temporal_filter
3. extract_author_refs

```python
for m in _AUTHOR_YEAR_RE.finditer(full_text):
    # Groups: (1=B1-B3 author, 2=B1-B3 year, 3=B4 author, 4=B4 year)
    if m.group(1) and m.group(2):       # B1-B3: (Author YYYY)
        author = m.group(1).split()[0]
        year   = int(m.group(2))
        fmt    = "B1-B3"
    elif m.group(3) and m.group(4):     # B4: Author (YYYY)
        author = m.group(3).split()[0]
        year   = int(m.group(4))
        fmt    = "B4"
    else:
        continue
```

```python
# Expand comma-separated brackets: [7,8] -> ref_numbers [7, 8]
all_hits = []  # (ref_number, match_start, match_end)
for m in matches:
    numbers = [int(n.strip()) for n in m.group(1).split(",")]
    for n in numbers:
        all_hits.append((n, m.start(), m.end()))
```

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Pipeline:

4. count_refs: check the citation we found with gold docs number
5. compute_scores:
    - Layer 1: TF - IDF (sentence - corpus)
    - Layer 2: Dense (sentence - corpus)
    - Layer 3: Previous Dense Result (considering the full text)
    - Weight: 0.4 - 0.4 - 0.2

- Result: 0.53

- Why and what we can do if have more time:

- Over-extraction Noise (locate 80 references but only 4 gold doc)
- Insufficient Context Window for Semantic Mapping (short sentence with several references)
- ……

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Pipeline — Scoring and Results

4. `count_refs`: check the citations found against gold document numbers
5. `compute_scores`:
   - Layer 1: TF-IDF (sentence → corpus)
   - Layer 2: Dense (sentence → corpus)
   - Layer 3: Previous Dense Result (full text)
   - Weights: 0.4 – 0.4 – 0.2

Result: NDCG@10 = 0.53

Limitations (given more time):

- **Over-extraction noise** — located ∼80 references but only 4 are gold documents
- **Insufficient context window** — short citation sentence with multiple references loses semantic meaning

BM25 using BM25Okapi

Model	NDCG@10	MAP	Recall@100
TF-IDF	0.4841	0.3804	0.7512
TA Default	0.4555	0.3473	0.7154
TA Tuned	0.4630	0.3561	0.7251
Body Default	0.4843	0.3837	0.7678
Body Tuned	0.5171	0.4171	0.7961

Best Result: Body Tuned ($k_1 = 2.0,\ b = 0.9$) achieved highest NDCG@10 and MAP.

BM25 using CountVectorizer

Model	NDCG@10	MAP	Recall@100
TF-IDF	0.4841	0.3804	0.7512
BM25 TA Default	0.4555	0.3473	0.7154
BM25 TA Tuned	0.4630	0.3561	0.7251
BM25 Body Default	0.4843	0.3837	0.7678
BM25 Body Tuned	0.5171	0.4171	0.7961

Best Result: Body Tuned $(k_1 = 2.0,\ b = 0.9)$ outperformed TF-IDF baseline.

BM25 Findings

- Full body text retrieval outperforms title+abstract only.
- Best body params: $\{k_1 = 2.0,\ b = 0.9\}$
- Best TA params: $\{k_1 = 1.8,\ b = 0.9\}$
- Both BM25Okapi and CountVectorizer implementations agree on best configuration.
- **Best overall BM25:** Body Tuned $\rightarrow$ NDCG@10 $= 0.5171$, MAP $= 0.4171$

Dense Retrieval Models Comparison

Model	NDCG@10	MAP	Recall@100
SPECTER2	0.4193	0.3190	0.7148
all-MiniLM-L6-v2	0.5073	0.4080	0.8104

Retrieval performance comparison across **dense embedding models**.

Key Findings

- all-MiniLM-L6-v2 achieved the best performance across all metrics.
- Higher Recall@100 indicates stronger retrieval coverage.
- Dense MiniLM improved over TF-IDF (0.507 vs. 0.484), but BM25 with body text (0.517) remained the stronger single model — real gains only came from combining both.

8-Signal Hybrid Retrieval System

- We built an **8-signal retrieval system** combining lexical, semantic, and metadata signals.
- Signals used:
  - **Neural:** Dense MiniLM embeddings (cosine similarity)
  - **Neural:** Dense BGE-small embeddings (diversity)
  - **Keyword:** BM25 over body text — full query ($k_1 = 3.0,\ b = 0.95$)
  - **Keyword:** BM25 over full text (TA + body) — full query ($k_1 = 3.0,\ b = 0.85$)
  - **Keyword:** TF-IDF cosine on full corpus — TA query
  - **Keyword:** TF-IDF cosine on full corpus — full query
  - **Metadata:** Domain match (binary — same research domain)
  - **Metadata:** Venue match (binary — same publication venue)
- Each signal is normalized to $[0, 1]$ row-wise before fusion.

Signal Weight Optimization (Coordinate Descent)

Weights optimized by iteratively updating one weight at a time until convergence.

Signal	**Weight**	**Type**
Domain Match	0.237	Metadata
Dense MiniLM	0.236	Neural
Dense BGE-small	0.151	Neural
TF-IDF Full (full query)	0.119	Keyword
TF-IDF Full (TA query)	0.114	Keyword
BM25 Body (full query)	0.091	Keyword
Venue Match	0.047	Metadata
BM25 Full (full query)	0.006	Keyword

Key insight: Domain Match (metadata) is the strongest single signal — outweighs even neural models.

Final Hybrid Retrieval Results

Model	NDCG@10	MAP	Recall@100
TF-IDF (TA)	0.4841	0.3804	0.7512
Dense MiniLM	0.5073	0.4080	0.8104
3-Way Hybrid	0.5916	0.4774	0.8502
8-Signal Fusion	**0.7373**	**0.6461**	**0.9178**

3-Way Hybrid: Dense (0.6) + BM25 Body (0.3) + TF-IDF (0.1)

Key Findings

- The 8-signal optimized fusion achieves **+52% NDCG@10** over TF-IDF baseline.
- Recall@100 = 0.9178 —¿ nearly all relevant papers are in the top-100.
- MRR@10 = 0.874 —¿ the first correct result appears at position ≈ 1.1 on average.

Per-Domain Performance (8-Signal Fusion)

Domain	NDCG@10	MAP	R@10	n
History / Political Sci.	1.000	1.000	1.000	1 each
Geology	0.957	0.896	1.000	2
Psychology	0.894	0.833	0.917	6
Materials Science	0.861	0.768	0.710	5
Chemistry	0.820	0.679	0.664	10
Physics	0.765	0.665	0.722	6
Biology	0.713	0.609	0.675	21
Medicine	0.687	0.619	0.713	21
Computer Science	0.553	0.466	0.404	12
Philosophy	0.237	0.111	0.500	1

Note: Philosophy scores very low — likely due to highly abstract, broad query terms with little domain overlap.

What We Learned (1/2)

Surprises

- Domain Match was the **single strongest signal** (weight 0.237), higher than both neural models.
- BM25 peaked at unusually high params ($k_1 = 2.0,\ b = 0.9$), far above common defaults ($k_1 = 1.2,\ b = 0.75$).
- Adding metadata alone lifted NDCG@10 by over **+14 points** on top of a strong hybrid.

Insights

- Lexical, neural, and metadata signals capture **different aspects** of relevance - none alone is sufficient.
- Full body text consistently outperforms title+abstract across **every** model type.
- Two neural encoders together outperform either alone.

What We Learned (2/2)

Lessons

- **Never skip metadata**: domain/venue signals are cheap to compute and surprisingly powerful.
- **Normalize before fusion**: row-wise $[0, 1]$ normalization is essential before combining signals.
- **Tune hyperparameters**: BM25 default vs. tuned gave a consistent $+5$–8 point improvement.
- More **diverse** signals $>$ one perfect signal.

Hybrid Retrieval System Second approach

- **Input:** Query paper (title + abstract + full_text)
- **10 signals** combined: Dense (MiniLM, SPECTER2 base/prox, GTE, SciBERT) + Lexical (TF-IDF, BM25 on TA & full_text) + Meta (domain, year, citation extraction)
- Each signal **min-max normalized per query**, then fused via weighted sum $\sum_i w_i \cdot s_i$
- **Weights learned** via 5-fold CV + differential evolution $\rightarrow$ top-100 per query submitted

Iteration & Results

Addition	**NDCG@10**
TF-IDF baseline	0.4841
MiniLM dense baseline	0.5073
6-signal hand-tuned hybrid	0.6053
+ extended TF-IDF on full_text	0.6156
+ SPECTER2 on full_text, Nelder-Mead	0.6665
10 signals + differential evolution	0.6940
+ GTE + BM25 + 5-fold CV	**0.6987**
+ SPECTER2-prox-FT + 15-vector ensemble	0.6970

Key Techniques Explained

- **SPECTER2-prox**: BERT variant with a *proximity adapter* fine-tuned on citation prediction: citing & cited papers are pulled close in embedding space. Directly aligned with our task.
- **GTE**: general-purpose sentence encoder (Alibaba). Adds semantic diversity beyond scientific-domain models.
- **Nelder-Mead**: *local* gradient-free optimizer; fast but gets stuck in local minima.
- **Differential evolution**: *global* population-based optimizer; explores the full weight space. Replaced Nelder (+0.03 NDCG@10).
- **5-fold cross-validation**: split 100 public queries into 5 folds; train weights on 4, validate on 1, average. Prevents overfitting to the public set.
- **15-vector ensemble**: random seeds $\times$ 5 folds $=$ 15 weight vectors averaged. Reduces variance for more stable held-out performance.

Error Analysis — Key Findings

Score Distribution (100 queries):

- 37 queries scored ≥ 0.90 — system works well for most queries
- 18 queries scored < 0.50, and **3 scored exactly 0.00**

Main Finding — Ranking Problem, Not Retrieval:

- **17 out of 18** failures: relevant papers are in top-100 but not reaching top-10
- Only **1 out of 18**: relevant papers missing from top-100 entirely

Root Cause — Domain Signal Too Broad:

- Domain match correctly identifies the field but fills top-10 with the **wrong sub-topic**
- *Example:* CardioID (biometric device pairing) $\rightarrow$ system returned security & IoT papers, both tagged as Computer Science

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